

The SMOS mission: A New Eye on Solar Radio Emissions

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Introduction

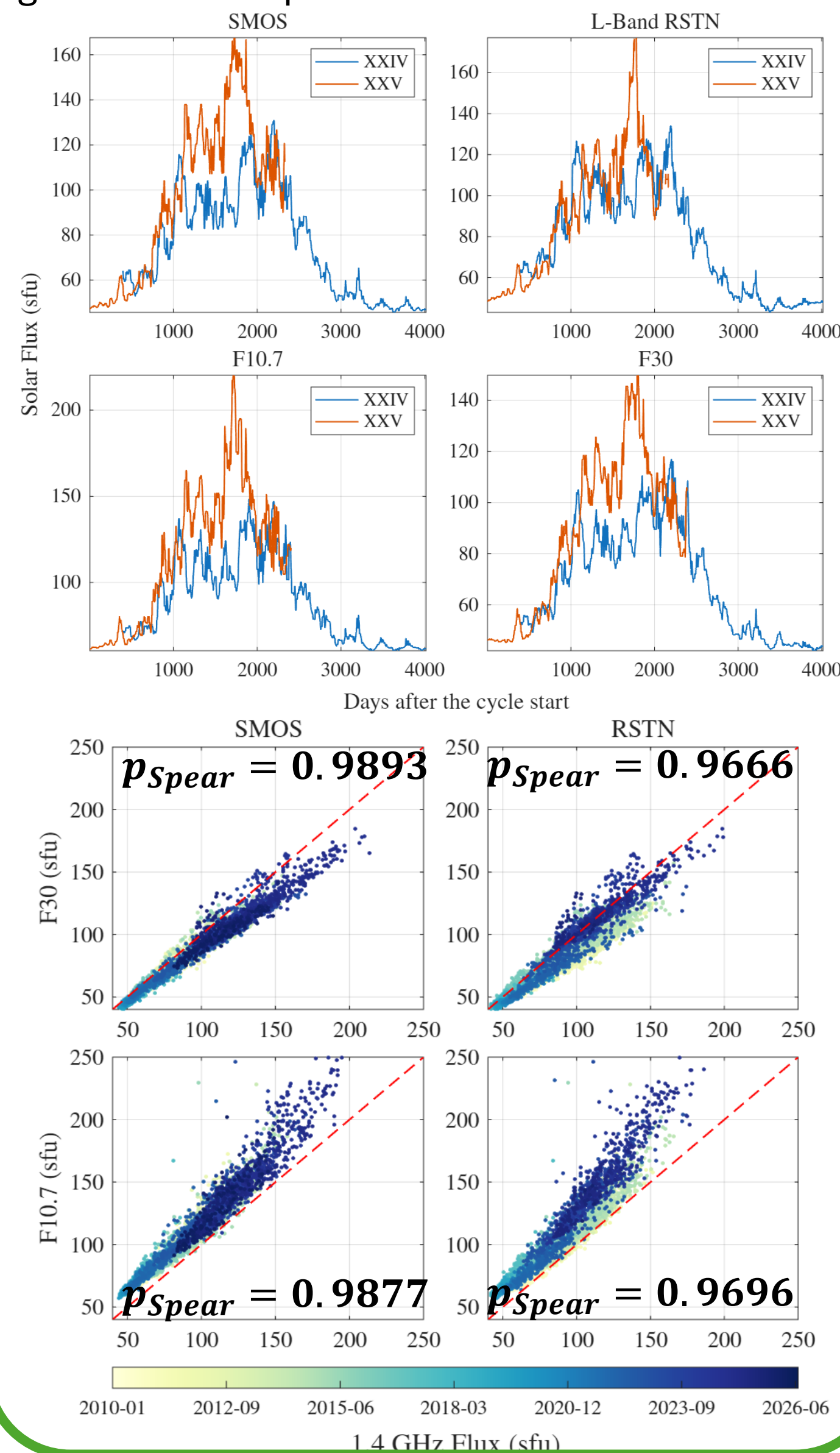
The Soil Moisture and Ocean Salinity (SMOS) mission, launched in 2009, enables long-term solar monitoring using the Microwave Imaging Radiometer with Aperture Synthesis (MIRAS), which performs full-polarization measurements at **1.41 GHz (21.2 cm)** with a temporal resolution of **1.2 s**. Previous studies have demonstrated the **high quality and long-term stability** of SMOS solar flux measurements.

Thanks to its high temporal resolution and continuous coverage, SMOS is particularly suitable for **solar burst detection**, allowing the characterization of both flux intensity and the **Degree of Circular Polarization (DoCP)** with **potential near real-time** capabilities. This work presents the SMOS L-band solar flux, explores its relationship with solar activity and ionospheric parameters, and highlights the potential of SMOS as a **reliable source of information for heliophysics, ionospheric modelling, and space weather operations**.



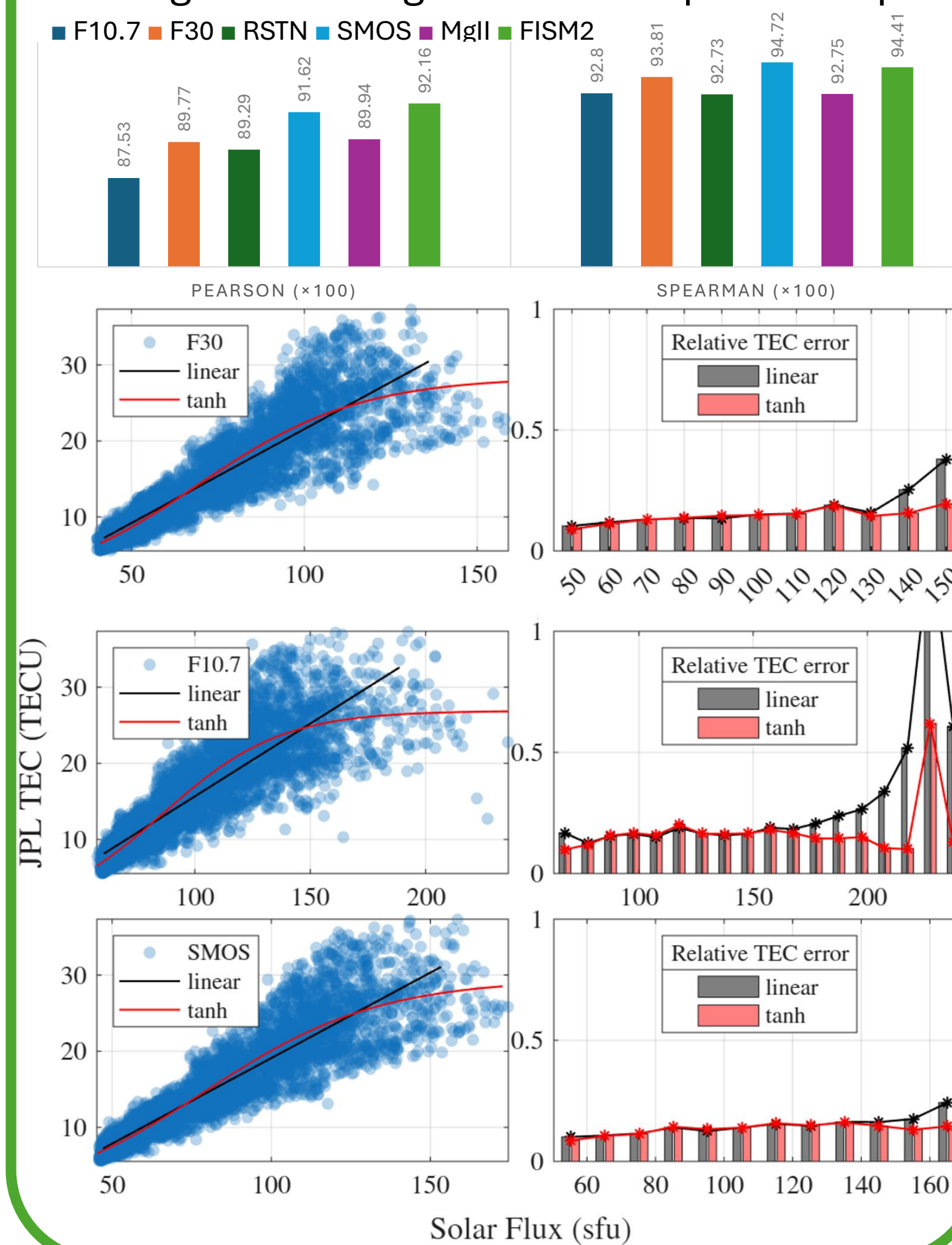
Data Quality

- The daily solar flux derived from SMOS shows **greater long-term stability than RSTN** measurements at the same frequency. In recent years, RSTN data exhibit a clear flux underestimation, evident both in the shape of the latest solar cycle and from frequent cases where the daily flux falls below the F30.
- The **uncertainty** at daily scale is consistently **below 5%**, similar if not slightly lower than the ground telescopes.



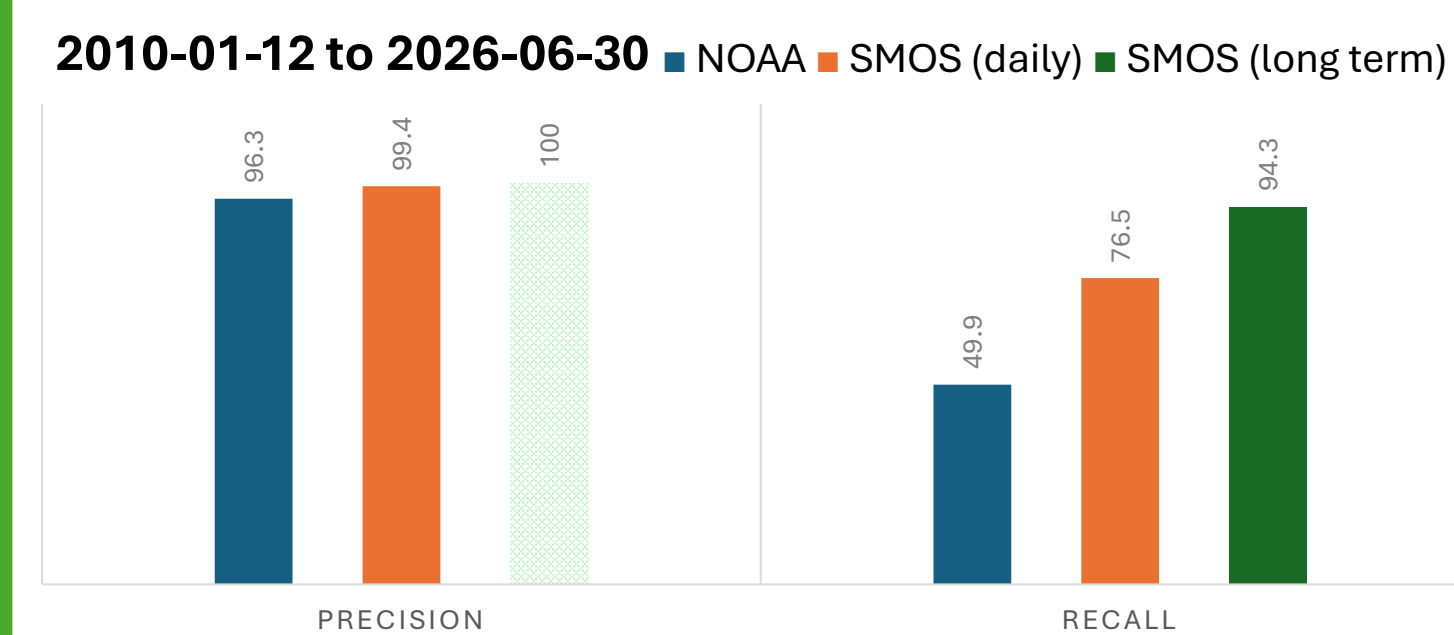
Solar Activity Proxy

- Correlation with ionospheric parameters (e.g., TEC and foF2) has been assessed. **SMOS consistently outperform both F10.7 and F30** in describing ionospheric variability, showing higher correlation, stronger linearity, lower saturation and RMSE, and greater stability across solar cycles and ASC/DES phases.
- Due to its data quality, SMOS is **competitive with UV/EUV proxies** (e.g., MgII, FISM2).
- Preliminary **results from foF2** measurements at multiple stations are broadly **consistent with TEC-based findings**, although correlations are less structured, likely because local and seasonal effects are not averaged out, unlike in the TEC dataset, which is based on the global average of JPL ionospheric maps.

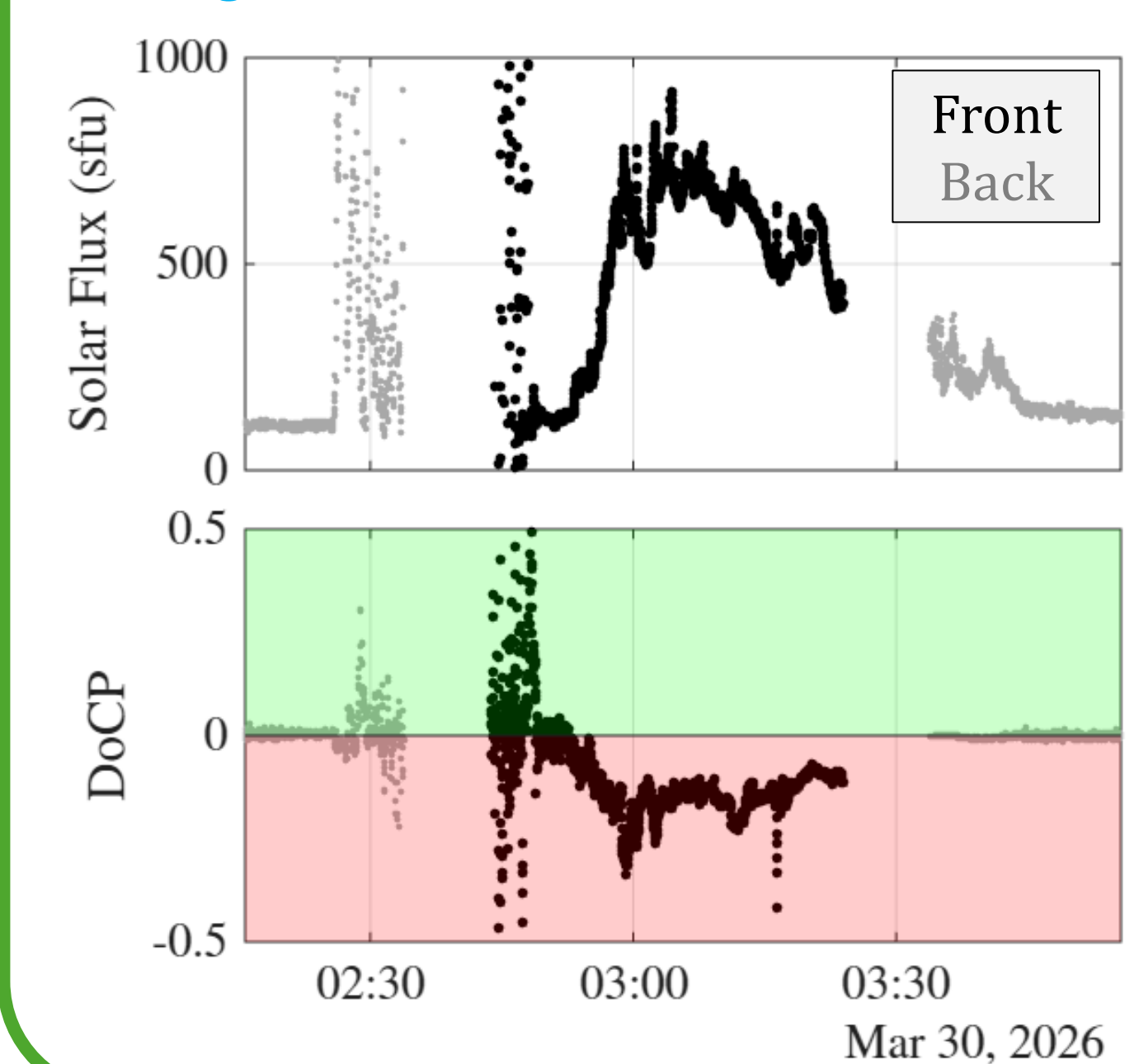


Solar Bursts Detection

- Solar radio bursts are automatically detected using a dedicated algorithm combining **statistical characterization** of 1.4 GHz events and **Random Forest classification** to reject RFI and orbital biases, running **daily** on the antenna's main lobe acquisitions. The database is **consolidated monthly** by accounting for the high time resolution data from multiple sources.
- Validation against the NOAA's archive shows similarly **high precision**, while SMOS achieves **>50% higher recall** and an **estimated long-term completeness exceeding 94%**.



- Note that **also some prominent events are missing** from the NOAA database.



The solar flux from the SMOS mission has high **data quality, long-term stability**, and the possibility of **near-continuous monitoring and potential for near real-time** operation. These properties give SMOS enormous **potential for both the heliophysics and space weather communities**.

These properties make it a valuable **proxy for solar activity and ionospheric variability**, with performances that surpass the individual use of traditional indices such as F10.7 and F30. Analyses on **foF2**, regional **TEC**, and seasonal components further highlight **marginal improvements driven by the usage of SMOS**.

The quality and temporal coverage of the observations enable **monitoring of microwave bursts, including polarization information**, that are highly competitive with existing ground and space-based alternatives. **Circularly polarized bursts** at SMOS' frequency are of interest for key technologies like **GNSS**.

Space weather is one of the pillars of the SMOS Data Innovation Science Cluster (SMOS-DISC), an ESA project launched in March 2026. Through SMOS-DISC, ESA fosters engagement with solar scientists and operational space weather stakeholders. **A SMOS-DISC-sponsored open science call is expected by the end of the year** to further explore the SMOS dataset for space weather applications, support continued improvements in solar signal estimation and calibration, and advance the development of additional products such as SMOS-derived TEC.

For further information, please contact the relevant points of contact provided!

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